Experimental Contact V: The Granularity Ceiling and the Emulation Programme

Where Finiteness Could Bite Quantum Computation, and How to Run FRC on Hardware

Yosef Akhtman

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Abstract

This note closes the last two items of the experimental tier list. *The granularity ceiling (T8):* ledger denominators are intrinsic — $1/\sqrt{2}$ has exact denominator 2 in every cyclotomic ledger, so k coherent two-way splittings carry minimal integer-tally norm 2^k — and the counting Born rule is derived only on the sub-horizon class, where every outcome probability is a multiple of $1/W > 1/p$. The representable coherent-splitting depth in a carrier $\mathbb{F}_p$ is therefore exactly $d^* = \lfloor \log_2 p \rfloor$, and single-shot probability resolution finer than $1/p$ lies outside the derived regime, with deviations quantised in multiples of p. With the corpus carrier the ceiling sits at $d^* \approx 405$ splitting levels (window $\Omega \sim 10^{122}$) or ≈ 202 (window $\sqrt{\Omega}$) — exactly located, and unobservably remote, since fault-tolerant algorithms read their answers through $O(1)$-probability events. The honest conclusion is a survival statement: FRC predicts quantum computation works exactly as standard quantum mechanics says at every accessible depth, and the first deviation, if it exists, is quantised and lives beyond any practical resolution. *The emulation programme (T9):* the forced-basis measurement of the formalism compiles to a shallow circuit — index the object cycle by (outcome, core) registers, apply an inverse quarter-turn QFT on the core register, read both registers. The 157/53/13 example becomes a 12-level circuit ($I_3 \otimes \mathrm{QFT}_4^\dagger$, trapped-ion qudit or 4-qubit embedding) and the Tsirelson configuration 641/41/17 a 16-level circuit ($I_2 \otimes \mathrm{QFT}_8^\dagger$); a hardware run certifies the selection rules, the uniform-outcome law for windings, the three-outcome Malus law, and the $\pi/40$ setting granularity of the composite gate. All claims are machine-verified (`granularity.py`, `emulation.py`; twelve checks, all passing).

1 T8: the granularity ceiling

Proposition 1.1 (Denominator and tally growth). *In the ledger $\mathbb{Q}(\zeta_8)$ the element $1/\sqrt{2} = (\zeta_8 + \zeta_8^{-1})/2$ has exact denominator 2, and the prime 2 is ramified: no change of cyclotomic ledger removes it. A state prepared by k coherent two-way splittings has minimal denominator exactly $2^{\lceil k/2 \rceil}$ and minimal integer-tally norm 2^k (k even; 2^{k+1} odd).* Verified exactly for $k = 1 \ldots 12$, including minimality of the integerised gcd (`granularity.py`, G1–G2).

Theorem 1.2 (The ceiling). *The counting Born rule of the companion [1] holds on the sub-horizon class, total tally $W < p$, where every outcome probability is a multiple of $1/W$. Hence: (i) the representable coherent-splitting depth in a carrier $\mathbb{F}_p$ is $d^* = \lfloor \log_2 p \rfloor$ exactly; (ii) single-shot probability resolution finer than $1/p$ is outside the derived regime; (iii) past d^* the window lift is ambiguous and the competing Born assignments differ by multiples of p — the same wrap quantisation as the Sorkin analysis [2].* Verified in the toy carrier $\mathbb{F}_{641}$: $d^* = 9$, unique lifts through depth 9, demonstrably distinct Born ratios from the two valid lifts at depth 10, discrepancies in $p\mathbb{Z}$ (G3–G4).

Remark 1.3 (The honest confrontation)**.** With the corpus carrier the ceiling sits at $d^* \approx 405$ coherent splitting levels per amplitude path (window $\Omega \sim 10^{122}$), or ≈ 202 if the operative window is the coherence horizon $\sqrt{\Omega}$. Neither number is reachable: fault-tolerant algorithms — Shor's included — extract their answers through events of $O(1)$ probability and polynomially many repetitions, never through single-shot resolution of exponentially small probabilities. Two consequences, both survival-type. First, FRC predicts that quantum computation succeeds exactly as standard quantum mechanics prescribes at every accessible depth: the programme licenses no near-term "quantum computers will fail" claim, and any observed systematic failure of deep circuits is *not* explicable by FRC granularity. Second, the ceiling is nevertheless real, exactly located, and structurally specific — deviations, where they begin, are quantised in carrier multiples rather than gradual — so the framework remains falsifiable in principle on this axis, while honestly reporting that this axis will not be the one that decides it.

2 T9: the emulation programme

The worked examples of the corpus are small enough to run on existing hardware, and the forced-basis measurement compiles to a standard circuit.

Proposition 2.1 (Compilation)**.** *Index the object cycle C_{12} of the* 157/53/13 *example by (α, ℓ) with $u = g_{\mathcal{O}}^{\alpha+3\ell}$. The measurement vectors factor as $v_{r,\alpha} = \delta_\alpha \otimes (\mathrm{i}^{r\ell})$, so the forced-basis measurement is: apply $I_3 \otimes \mathrm{QFT}_4^\dagger$, read (α, r). The QFT_4 is the textbook two-qubit circuit (H–CS–H–SWAP,* verified entry-by-entry*); the outcome register is a qutrit or two qubits. The* 641/41/17 *Tsirelson configuration compiles identically as $I_2 \otimes \mathrm{QFT}_8^\dagger$ on* 16 *levels.* Unitarity and the exact correspondence of circuit columns to forced-basis vectors verified (`emulation.py`, M1, M4, M5).

Proposition 2.2 (What a run certifies)**.** *The compiled circuits reproduce, to 10^{-12} against the exact ledger predictions: the selection rule and uniform-outcome law for all twelve winding states ($P(\alpha) = \frac{1}{3}$ in the matched channel, zero elsewhere); the three-outcome Malus law $P(\alpha) = |1 + \lambda\,\omega^\alpha|^2/6$ for doublet states across the full phase sweep; and, in the* 641 *compilation, the σ_x doublet law $w_\pm = |1 \pm \lambda|^2/4$ over the $\pi/40$ setting granularity of the composite gate [3].* (M2, M3, M5.)

Remark 2.3 (Experiment cards)**.** *Card 1 (157/53/13):* twelve levels; prepare winding states (phase ramps) or doublets; apply $I_3 \otimes \mathrm{QFT}_4^\dagger$; read both registers; certify selection rules, uniformity, Malus. Platforms: trapped-ion qudits ($d = 12$ native) or a four-qubit embedding using 12 of 16 levels. *Card 2 (641/41/17):* sixteen levels; $I_2 \otimes \mathrm{QFT}_8^\dagger$; certifies the Malus law at $\pi/40$ and, with two copies and the offset readout of the network note [4], the full Tsirelson-saturating CHSH statistics. The emulation does not test nature — it tests the formalism's statistics on hardware, provides the community a concrete handle on the framework, and exercises precisely the circuits whose large-carrier limits carry the programme's physical claims.

3 Status: the tier list closes

T1 through T9 are now closed: Sorkin nullity and wrap quantisation; boost-dispersion exactness with the finite spinor double cover; the composite gate with exact Tsirelson saturation; the network game at the complex-quantum table; the BMV mechanism and forward prediction; the forbidden-collapse theorem and the dilation floor with exact revivals; the exact equivalence-principle null and the two scaling laws; the granularity ceiling, exactly located and honestly remote; and the emulation programme, compiled and verified. Eight validation suites, sixty-six exact checks, all passing; five companion notes filed.

What remains is the consolidation: the Ω *ledger* — the single table on which every entry above depends on one carrier scale, where the Sorkin window, the dispersion exactness, the Born readout, the decoherence floor, the granularity ceiling, the RH decidability window, and the gravity paper's acceleration floor must all be satisfied by the same Ω. Assembling that table, and keeping it current as bounds improve, is the standing experimental obligation of the programme.

Reproducibility

The scripts `granularity.py` and `emulation.py` reside in the `validation/` folder of the companion draft, slated for https://github.com/gamayos/frc-quantum-numerics. Recorded output: twelve `PASS` lines, two `INFO` experiment cards, and the terminal lines `granularity: all checks passed`, `emulation:  all checks passed`.

References

[1] Y. Akhtman. *Quantum Observation in Finite Ring Cosmology.* Companion draft, June 2026.

[2] Y. Akhtman. *Experimental Contact I: Exact Nullity Tests.* Companion note, June 2026.

[3] Y. Akhtman. *Experimental Contact II: The Composite Gate.* Companion note, June 2026.

[4] Y. Akhtman. *Experimental Contact III: The Network Game and the Gravitational Channel.* Companion note, June 2026.

[5] Y. Akhtman. *Experimental Contact IV: The Decoherence Floor and the Equivalence Principle.* Companion note, June 2026.